

Radiolytic stability of paracetamol in montmorillonite complexes: Influence of the mineral matrix on radiolytic yields and molecular persistence

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ABSTRACT

The radiolytic fate of pharmaceuticals in terrestrial environments could be significantly modulated by the inorganic fraction of soils, which could act as a reservoir for contaminants. This study investigates the radiation stability of paracetamol using montmorillonite as an important soil mineral. By integrating spectroscopic (XRD, ATR-FTIR), chromatographic (ESI-HPLC-MS), and thermal evidence (DSC/TGA), the results demonstrate that while paracetamol in the bulk aqueous phase undergoes rapid radiolytic transformation at low doses (≤ 1.5 kGy), its association with the mineral matrix significantly suppresses degradation efficiency without altering the chemical pathway. Notably, in the solid-state dried complex, paracetamol remained chemically intact and showed no significant degradation even after exposure to the maximum absorbed dose of 12.4 kGy. This molecular confinement induces a significant reduction in radical accessibility, decreasing the initial radiolytic yield (G_0) of paracetamol by approximately 65.5%, from 1.07 molecules/100 eV in aqueous solution to 0.37 molecules/100 eV in the water-clay system, and to practically zero in the dry complex. The identification of identical radiolytic products (hydroquinone, acetamide, and malonic acid) in both systems confirms that the mineral matrix inhibits breakdown through matrix-induced protection associated with the absorption of ionizing energy, evidenced by the attenuation of structural Si-O vibrations (~ 1000 cm⁻¹) and potentially likely competition kinetics for reactive $\bullet$ OH radicals. The chemical persistence of paracetamol was confirmed by its characteristic thermal signature at 169 °C in post-irradiated complexes. Ultimately, this work elucidates the role of clay minerals as protective environmental reservoirs that could promote the long-term persistence of adsorbed contaminants by suppressing their radiolytic breakdown.

1. Introduction

The persistent release of pharmaceutical compounds into aquatic and terrestrial environments (Maddela et al., 2022; Singh et al., 2025) has prompted the development of effective remediation technologies (Ortúzar et al., 2022). Among these, Advanced Oxidation Processes (AOPs) are prominent due to their capacity to mineralize recalcitrant pollutants (Attri et al., 2018; Bao et al., 2022a; Sayed et al., 2018). Paracetamol (acetaminophen) is an emerging contaminant of significant concern (Ayoub, 2021); its detection in water sources (Al-howri et al., 2024) and soils, at concentrations reaching 1400 μ g kg⁻¹, is associated with oxidative stress and endocrine disruption in various organisms (Guiloski et al., 2017). The soil environment represents a complex,

natural four-phase system (Tarbuck and Lutgens, 2005) that functions as a long-term reservoir where contaminants accumulate through surface runoff and the widespread practice of irrigation with reclaimed wastewater (Lees et al., 2016; Singh et al., 2025). While the radiolytic degradation of paracetamol in dilute aqueous solutions is well-documented (Szabó et al., 2012) demonstrating high efficiency mediated by hydroxyl radicals with a standard primary yield of $G(\bullet\text{OH}) = 0.28$ μ mol J⁻¹ (LaVerne, 2000) its behavior when interacting with the solid phases of the soil remains a critical knowledge gap. Ionizing radiation (IR) offers distinct advantages over conventional AOPs (Bao et al., 2022; Madureira et al., 2022), as it enables reagent-free degradation through the simultaneous generation of oxidative and reductive species (Spinks and Woods, 1990).

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Inorganic soil components, such as montmorillonite, possess high surface areas and reactive sites (edges and interlayer channels) that can sequester contaminants and modulate their reactivity (Allison, 1973; Yuan et al., 2006). Previous research suggests that solid surfaces play a dual role in radiation chemistry: they can act as reaction micro-environment modulators or as modulators of the reaction microenvironment (El-khatib et al., 2022; Khattari et al., 2023). Clay minerals can store energy from ionizing radiation and transfer it to adsorbed reactants (Lainé et al., 2016), a phenomenon observed in prebiotic chemistry (Ramos-Bernal and Negron-Mendoza, 1998) but less explored in environmental remediation. The moisture content in terrestrial matrices plays a pivotal role in radiolytic processes; as noted by McNamara et al. (2003) while gamma irradiation is an effective biocidal tool for soil sterilization, its impact on the matrix's chemical stability is significantly heightened by moisture due to the indirect action of water radiolysis products.

This study aims to provide fundamental insights into the stability of pharmaceuticals associated with the soil mineral fraction by evaluating how the inorganic framework dictates the radiolytic fate of paracetamol with and without water. To achieve this, three distinct configurations are evaluated: aqueous solutions, aqueous suspensions, and dried pre-adsorbed complexes, serving as surrogate models for fully hydrated and anhydrous soil analogues, respectively. By integrating spectroscopic (UV-Vis, ATR-FTIR, XRD), chromatographic (ESI-HPLC-MS), and thermal analysis (DSC/TGA), this work demonstrates how the aluminosilicate lattice induces a matrix-dependent inhibition of degradation. Ultimately, these findings contribute to the fundamental understanding of radiolytic processes involving the mineral fraction as a key modulator of contaminant persistence in terrestrial environments.

2. Experimental

2.1. Materials

All reagents were of analytical grade and used as received without further purification. Paracetamol (acetaminophen, $C_8H_9NO_2$, white crystalline powder, purity $\geq 99\%$), hexadecyl trimethyl ammonium bromide (HDTMABr, $C_{19}H_{42}BrN$, white powder, purity $\geq 99\%$), methanol (CH_3OH , liquid, purity $\geq 99.9\%$) and ethanol (C_2H_5OH , liquid, purity $\geq 99.5\%$) were purchased from Sigma-Aldrich (St. Louis, Missouri, USA). Deionized water (18.2 M Ω cm) was obtained from a Milli-Q Plus (Millipore, Burlington, Massachusetts, USA). Na-montmorillonite (SWy-2), a natural 2:1 phyllosilicate (solid powder), was sourced from the Clay Minerals Repository (University of Missouri, Columbia, Missouri, USA), and HDTMA-montmorillonite was prepared following the method described by Perezgasga et al. (2005). The organo-modified montmorillonite (HDTMA-Mnt) was synthesized using an ion-exchange process designed to achieve total interlayer channel blocking. Briefly, 10 g of Na-montmorillonite was reacted with 100 mL of an ethanolic solution containing 3.63 g of hexadecyltrimethylammonium bromide (HDTMABr), a concentration calculated to satisfy 100% of the clay's cation exchange capacity (101 meq/100g). The mixture was agitated at 100 rpm for 24 h, followed by three successive washing steps with deionized water and centrifugation at 20,000 rpm for 30 min. The resulting sediment was dried at 50 °C for 12 h, ground, and sieved to 125 μ m to ensure structural uniformity for the subsequent adsorption experiment. Both clays were characterized by X-ray diffraction (XRD), Attenuated Total Reflectance (ATR) Fourier transform infrared (FTIR) spectroscopy, and thermogravimetric analysis (TGA/DSC).

2.2. Analytical method

2.2.1. UV-Vis spectrophotometry

Absorbance measurements were performed to quantify paracetamol concentration in supernatants. A Varian Cary 100 Scan UV-Vis

spectrophotometer (Palo Alto, CA, USA), equipped with 1 cm quartz cells, was used at a fixed wavelength of 243 nm. Paracetamol concentrations for both adsorption and radiolysis experiments were calculated based on the Beer-Lambert Law. All experiments were performed in triplicate to ensure the reliability of the results ($n = 3$).

2.2.2. ATR-FTIR spectroscopy

To identify functional groups and surface interactions, solid-state spectra were acquired using a PerkinElmer Spectrum 100 spectrophotometer (Waltham, MA, USA) equipped with a Universal Attenuated Total Reflectance (ATR) accessory. This method is crucial for confirming the presence of paracetamol vibrations and the successful sorption in the clay.

2.2.3. X-ray diffraction (XRD)

The crystalline structure and interlayer spacing (d_{001}) were monitored using an Empyrean diffractometer (Malvern Panalytical, Almelo, Netherlands) with Cu-K α radiation (40 kV/40 mA). XRD is significant for distinguishing between edge-site adsorption and interlayer intercalation by measuring shifts in the (001) reflection.

2.2.4. DSC/TGA analysis

Simultaneous thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) were performed on a Netzsch STA 409 PC Luxx (Selb, Germany). This technique evaluates the thermal stability of the paracetamol-clay complex and confirms the organic load through weight loss stages after gamma radiolysis processes.

2.2.5. ESI-HPLC-MS chromatography

High-performance liquid chromatography coupled with mass spectrometry (HPLC-MS) was utilized to achieve the molecular specificity required for identifying radiolytic transformation products and quantifying paracetamol in complex matrices. Analyses were conducted on three types of samples: (1) paracetamol standard solutions, (2) supernatants from adsorption-desorption experiments, and (3) post-irradiation solutions. A Waters Alliance System (Milford, MA, USA), comprising a 515 HPLC pump coupled to an SQ-2 mass detector, was operated in negative electrospray ionization mode (ESI⁻). The optimized operational parameters included a capillary voltage of 1.46 kV, cone voltage of 23 V, source temperature of 350 °C, and a desolvation gas flow of 650 L h⁻¹. Chromatographic separation was achieved using a Waters Symmetry C18 column (75 mm $\times$ 4.6 mm, 3.5- μ m spherical particles). The mobile phase consisted of an isocratic elution of 75:25 water (H_2O): methanol (CH_3OH) (v/v) at a constant flow rate of 0.4 mL min⁻¹. Manual injections using a 20 μ L loop yielded consistent and reproducible detection of the deprotonated paracetamol ion [M-H]⁻ (m/z 150), with a characteristic retention time of 3.2 min. This technique is significant for distinguishing paracetamol from its primary radiolytic transformation products, which share overlapping UV-Vis absorption features. All chromatographic analyses were performed in triplicate to ensure the reliability of the results ($n = 3$).

2.2.6. Irradiation facilities and dosimetry

Samples were exposed to gamma radiation utilizing a ⁶⁰Co Gamma-beam 651-PT facility (Nordion, Ottawa, ON, Canada) located at the Instituto de Ciencias Nucleares, UNAM. The dose rate was maintained at a constant value of 180 Gy min⁻¹. To ensure high analytical purity and prevent experimental errors caused by organic residues, borosilicate glass tubes were pretreated by immersion in a warm (1:1) mixture of concentrated nitric acid (HNO_3) and sulfuric acid (H_2SO_4), followed by rinsing with deionized water and heating in an oven at 300 °C overnight. This cleaning protocol follows the established methodology of Draganić and Draganić (1971) to minimize contamination in irradiated systems. The dose rate for the specific irradiation positions was evaluated using Fricke dosimetry (ferrous ammonium sulfate-cupric sulfate solution). Based on these measurements, the total absorbed dose was determined

by calculating the required exposure times for each sample. All procedures and chemical yields utilized for these dose-rate estimations and subsequent dose calculations were performed in accordance with the protocols described by Spinks and Woods (1990).

2.3. Methods

2.3.1. Adsorption and pH studies

Batch adsorption experiments were conducted to evaluate the affinity of paracetamol for the mineral surfaces and to determine the influence of hydrogen ion concentration on the sequestration process. Suspensions were prepared by combining 6 mL of a paracetamol solution ($15 \mu\text{g mL}^{-1}$) with 200 mg of clay in 15 mL polypropylene centrifuge tubes. To investigate the effects of pH, parallel experiments were performed with solutions pre-adjusted to pH 5 using hydrochloric acid (HCl, 0.1 mol L^{-1}). This specific pH was selected to ensure that paracetamol remained in its neutral form ($\text{pK}_a = 9.38$), thereby minimizing electrostatic repulsion with the clay surface. The suspensions were agitated at 300 rpm for predetermined intervals (0.5 to 1440 min). Following agitation, samples were centrifuged at 20,000 rpm for 30 min, and the equilibrium pH was recorded. Adsorption efficiency was quantified by comparing the initial and residual concentrations via UV-Vis spectroscopy ($\lambda = 243 \text{ nm}$) and ESI-HPLC-MS. The resulting clay-paracetamol sediments were collected and dried at 50°C for 24 h for subsequent solid-state analysis.

To investigate the relationship between adsorption efficiency and the specific hosting site of paracetamol within the montmorillonite matrix, interaction experiments were conducted using HDTMA-modified montmorillonite as a diagnostic tool. The bulky HDTMA cations block the interlaminar channels, effectively inhibiting the sorption of molecules into the interlayer space.

2.3.2. Statistical analysis

All experiments were conducted in independent triplicates ($n = 3$), and the results are presented as mean values. An estimated relative experimental error of 5% was considered for the data points. Statistical significance between Na-montmorillonite and HDTMA-montmorillonite adsorption was evaluated using an independent t -test with a significance level of $\alpha = 0.05$.

For radiolytic yields (G-values) and analytical calibration curves, linear regression parameters were determined at a 95% confidence level. These parameters were calculated along with their respective standard deviation (SD) and coefficients of determination (R^2) to ensure the robustness of the results. Statistical calculations and data processing were performed using Python within a Google Colab environment, relying on the SciPy and NumPy libraries for regression analysis and error propagation. The specific mathematical approach applied to obtain these yields is further detailed in Section 2.3.6 Determination of radiolytic yields.

2.3.3. Alkaline stability tests

To examine the strength and reversibility of the paracetamol-clay interactions, dried Na-montmorillonite-paracetamol sediments (200 mg) were subjected to alkaline treatment. This was performed by resuspension in 6 mL of potassium hydroxide (KOH, pH 12). This extreme pH condition was chosen to promote electrostatic repulsion, as both the clay edges and the paracetamol molecules (in their phenolate form) acquire negative charges under these conditions. The mixtures were agitated at 300 rpm for 30, 60, 90, and 120 min, followed by centrifugation at 20,000 rpm for 30 min. The release of paracetamol into the supernatant was monitored via UV-Vis and ESI-HPLC-MS to evaluate the stability of the complexes against desorption.

2.3.4. Ionizing radiation experiments

To compare the radiolytic degradation pathways across different physical states, three distinct systems were prepared in pre-cleaned glass

vials. The systems included:

Aqueous paracetamol: 6 mL of air-saturated paracetamol solutions ($15 \mu\text{g mL}^{-1}$) to evaluate conventional radical-mediated degradation.

1. Paracetamol and clay suspension systems: 200 mg of Na-montmorillonite mixed with 6 mL of paracetamol solution ($15 \mu\text{g mL}^{-1}$, pH 5). These were pre-equilibrated for 120 min before irradiation to ensure maximum edge-site binding. This system was designed to study heterogeneous radiolysis and the influence of the mineral-water interface on radical availability.
2. Dried pre-adsorbed paracetamol-clay systems: Irradiation of 200 mg of dried sediments to investigate direct radiation effects and the aluminosilicate lattice modulates the interface chemistry, where ionizing energy is associated with localized framework alterations rather than the breakdown of the adsorbed pharmaceutical.

2.3.5. Thermal characterization

The post-irradiation thermal stability of the organic-clay complexes was assessed via simultaneous DSC/TGA using a Netzsch STA 409 PC Luxx. Samples (10 mg) were heated from 25 to 900°C at a rate of $10^\circ\text{C min}^{-1}$ under a controlled N_2 atmosphere (50 mL min^{-1}). This protocol enables the systematic differentiation of weight loss stages, specifically surface dehydration, the decomposition of adsorbed organic species (typically 150 – 400°C), and the final dihydroxylation of the aluminosilicate lattice at higher temperatures.

2.3.6. Determination of radiolytic yields

The effective radiolytic yields (G-values) for paracetamol degradation were calculated by correlating the concentration changes, determined via UV-Vis spectrophotometry, with the absorbed dose established through Fricke dosimetry (ferrous ammonium sulfate-cupric sulfate solution). These yields were expressed in the number of molecules destroyed per 100 eV of absorbed energy to allow for a direct comparison between the aqueous and heterogeneous systems.

The initial radiolytic yields of paracetamol ($G_0(\text{Para})$) were derived from the initial linear slope of the paracetamol concentration decrease as a function of the absorbed dose (evaluated over the range of 0.25 to 12.4 kGy). To accurately capture the initial reaction kinetics, only the first three experimental data points were used to determine the slope for the $G_0(\text{Para})$ calculation. Following the standardized calibration procedures and chemical yields described by Spinks and Woods (1990) and Cruz-González et al. (2016), $G_0(\text{Para})$ was calculated from the linear regression plot of the absolute number of destroyed molecules (N) versus the absorbed energy ($E_{100\text{eV}}$). This relationship for the initial linear stage of the degradation curve is mathematically expressed as:

$$G = dN / dE_{100\text{eV}}$$

Applying this slope method at the earliest stages of irradiation ensures a reliable estimation of the primary radiolytic efficiency. This approach guarantees that the calculated yields specifically reflect the primary degradation of the parent molecule mediated by water radiolysis products, effectively minimizing the interference of secondary recombination reactions or competition for radicals by degradation products.

3. Results and discussion

3.1. Characterization of aqueous systems

The paracetamol standard curve by UV-Vis spectrophotometry presents a coefficient of determination correlation greater than 0.9979. Additionally, the extinction coefficient calculated in this work was $9702.45 \text{ L mol}^{-1} \text{ cm}^{-1}$. Furthermore, the paracetamol standard curve obtained using ESI-HPLC-MS also presents a coefficient of correlation greater than 0.9994. The concentration used for all experiments was

15 $\mu\text{g mL}^{-1}$, as this concentration facilitates detection by the analytical techniques employed.

To ensure the molecular specificity required for the identification of paracetamol and its potential transformation products, an ESI-HPLC-MS method was validated. Fig. 1 presents the analytical characterization of the paracetamol standard. The chromatogram (Fig. 1, left) exhibits a defined peak at a retention time of 3.23 min under optimized isocratic conditions. This elution profile is coupled with the mass spectrum obtained in negative electrospray ionization mode (ESI⁻) (Fig. 1, right), which is dominated by the deprotonated molecular ion $[\text{M}-\text{H}]^-$ at $m/z = 150$. These parameters establish the analytical baseline used to confirm the chemical persistence of paracetamol and to distinguish it from transformation products in subsequent heterogeneous radiolytic experiments.

3.2. Characterization of solid systems

The XRD patterns of clays are presented in Fig. 2. The (001) reflection peak for Na-montmorillonite (black line) appears at $2\theta = 7.391^\circ$, corresponding to an interlayer spacing (d_{001}) of 11.95 Å. In contrast, HDTMA-montmorillonite (red line) exhibits a significant expansion of the interlayer distance ($d_{001} = 17.53$ Å), confirming the successful intercalation of the surfactant. This shift reflects the influence of the aliphatic hydrocarbon chains from the quaternary ammonium salt on the structural properties of the clay (Luo et al., 2016). HDTMA either successfully modified Na-montmorillonite or the organic HDTMA compounds were distributed across the surface of the material and interlaced between the layers of the clay mineral. Moreover, the ATR-FTIR spectrum in Fig. 3 shows the bands corresponding to the $-\text{CH}_3$ (2925, 1469, and 1380 cm^{-1}) and $-\text{CH}_2-$ (2850 and 1488 cm^{-1}) groups in the HDTMA molecule.

3.3. Adsorption and pH studies

Fig. 4 illustrates the time-dependent adsorption efficiency of paracetamol onto Na-montmorillonite at pH 5 and 9 over 1440 min (1 day). The largest adsorption (30%) occurred at pH 9 after 30 min of contact time, significantly lower than the 58.7% achieved at pH 5. This reduced efficiency at pH 9 (near the pK_a of paracetamol, 9.38, Fig. 5) results from electrostatic repulsion between deprotonated paracetamol molecules (phenolate form) and negatively charged clay edge sites ($-\text{O}-$ groups) (Schoonheydt et al., 2018). Both species acquire negative charges under

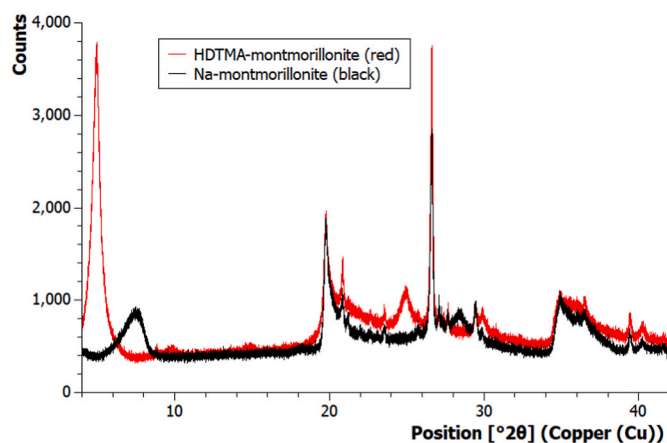


Fig. 2. XRD patterns of natural Na-montmorillonite (black) and HDTMA-modified montmorillonite (red). The expansion of the basal spacing from $d_{001}2\theta = 11.95$ Å to 17.53 Å confirms the successful intercalation of the quaternary ammonium salt within the silicate layers, creating a channel-blocked configuration. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

alkaline conditions, reducing adsorption.

To evaluate the effects of pH on paracetamol-clay interactions, experiments were conducted at pH 5, where speciation analysis (Fig. 5) confirmed that >95% of paracetamol remains neutral. This neutral form exhibited stronger affinity for clay surfaces, achieving 58.7% adsorption after 120 min, 95% increase compared to pH 9. The enhanced efficiency results from two synergistic factors: (1) at pH 5, <5% of paracetamol exists in deprotonated form (PhO^-), minimizing electrostatic repulsion, and (2) edge $-\text{OH}$ groups on montmorillonite remain protonated ($\equiv\text{Al}-\text{OH}_2^+$) below pH ~ 6.5 (Schoonheydt et al., 2018) encouraging hydrogen bonding and van der Waals interactions with neutral paracetamol molecules. These findings align with adsorption models for weak acid pollutants on aluminosilicates (Lawal et al., 2023), where optimal uptake occurs near $\text{pH} = \text{pK}_a$ of 4.

The possible interaction mechanisms between paracetamol and Na-montmorillonite could be $\pi-\pi$ and dipole-dipole interactions (Aydin et al., 2023). Bernal et al. (2017) reported the sorption of paracetamol on activated carbon, in which $\pi-\pi$ interactions occur due to the π electrons of the aromatic ring. However, these interactions occur

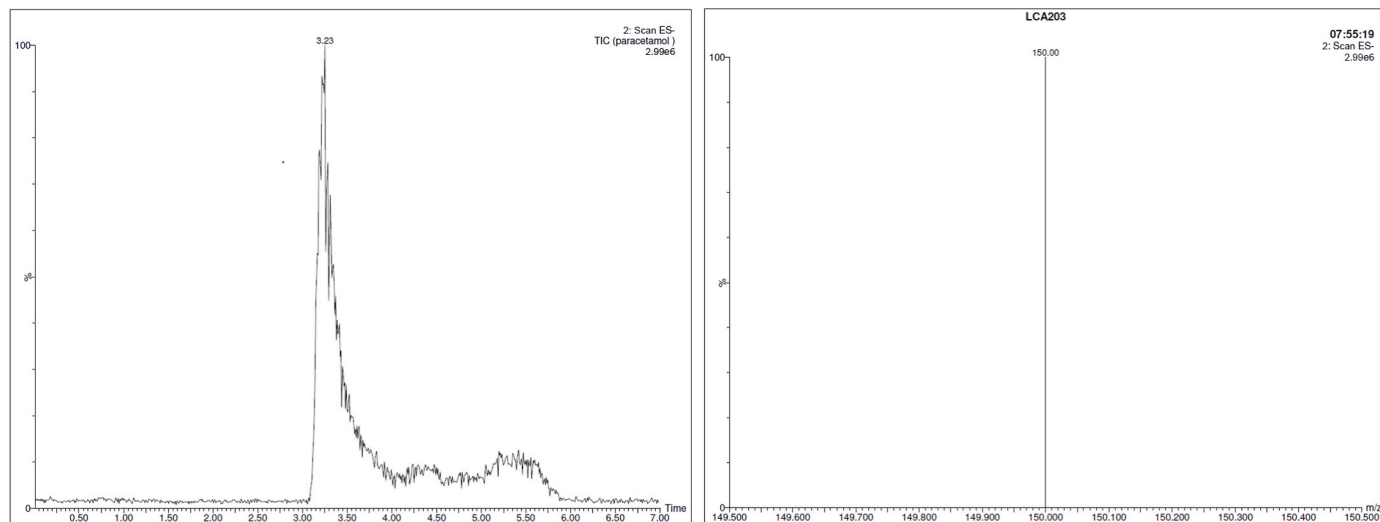


Fig. 1. HPLC-ESI-MS characterization of paracetamol standard solution ($15\text{ }\mu\text{g mL}^{-1}$). Left: Representative chromatogram showing the elution peak at a retention time of 3.23 min. Right: Negative-ion ESI mass spectrum highlighting the deprotonated molecular ion at $m/z = 150$. The combination of these signatures provides the diagnostic criteria for monitoring the structural integrity of the pharmaceutical in complex mineral matrices.

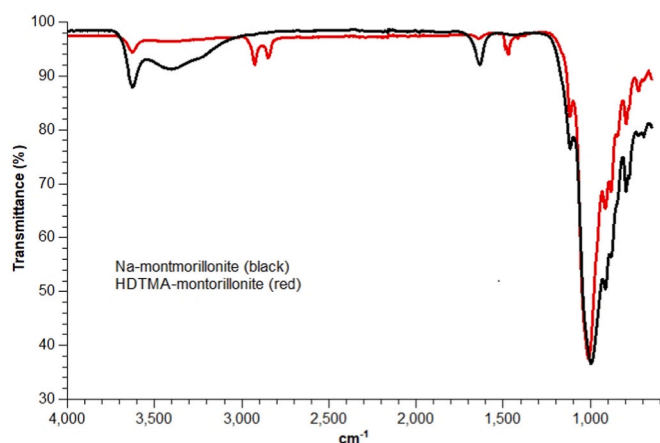


Fig. 3. ATR-FTIR spectra of Na-montmorillonite (black) and HDTMA-modified montmorillonite (red). The successful organic modification is confirmed by the appearance of characteristic aliphatic bands for $-\text{CH}_3$ (2925, 1469, and 1380 cm^{-1}) and $-\text{CH}_2-$ (2850 and 1488 cm^{-1}) groups of the surfactant molecule. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

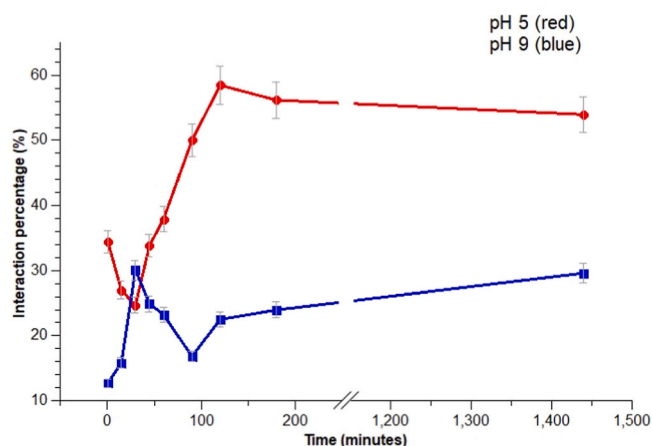


Fig. 4. Time-dependent adsorption efficiency of paracetamol on Na-montmorillonite at pH 5 (red and ●) and pH 9 (blue and ■). Optimal uptake (58.7%) is achieved at pH 5 after 120 min, where the neutral form of the drug minimizes electrostatic repulsion with the mineral edges, compared to the ~30% removal observed under alkaline conditions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

independently of the medium's pH. This can be confirmed by ATR-FTIR spectroscopy with the band at 1510 cm^{-1} (aromatic ring of paracetamol) shown in Fig. 6.

Fig. 7 shows the ATR-FTIR spectrum of the sediment from experiments at pH 5. The identified bands are similar to those of Na-montmorillonite; however, two bands at 2926 and 2854 cm^{-1} , corresponding to the methyl ($-\text{CH}_3$) group of paracetamol, indicate the successful interaction of paracetamol with the clay surface.

Table 1 provides a comprehensive summary of the ATR-FTIR spectroscopic observations, detailing the vibrational assignments and spectral features identified during both the ion-exchange modification of the clay and the subsequent paracetamol adsorption process.

To evaluate the adsorption efficiency and elucidate the specific hosting sites of paracetamol within the HDTMA-modified montmorillonite, experiments were performed under identical conditions (pH 5 and 120 min of contact time). The results yielded comparable adsorption efficiencies for both systems: 58.7% for Na-montmorillonite and 60.7%

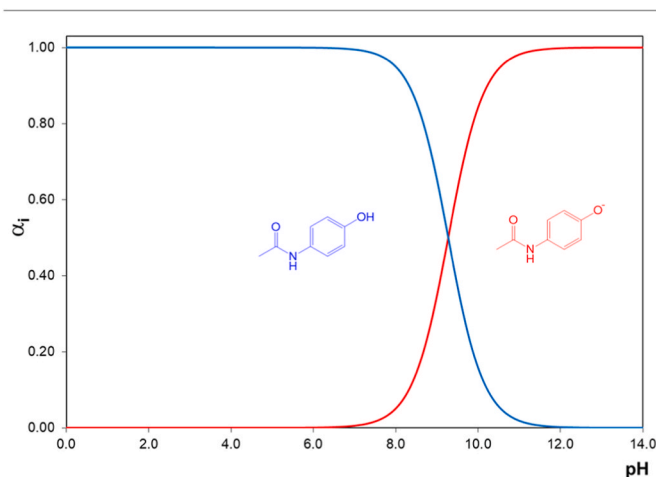


Fig. 5. Speciation distribution diagram for paracetamol as a function of pH. The intersection at $\text{pKa} = 9.38$ defines the transition between the neutral form (dominant at pH 5) and the anionic phenolate species, providing the theoretical basis for the enhanced adsorption affinity observed at acidic pH.

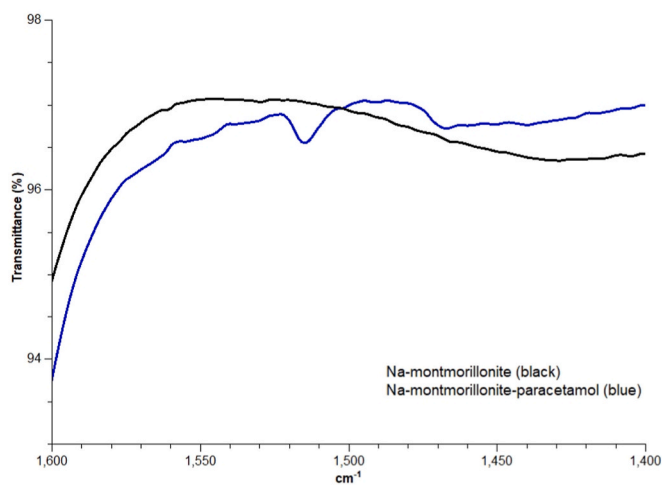


Fig. 6. Identification of the characteristic paracetamol aromatic ring vibration at 1510 cm^{-1} after adsorption onto Na-montmorillonite (blue line). The clear detection of this band confirms the presence of the pharmaceutical and suggests interactions between the π -electron system of the ring and the mineral edge sites. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

for HDTMA-montmorillonite (Fig. 8). These findings align with earlier investigations suggesting that van der Waals forces and aromatic ring interactions promote the coplanar orientation of paracetamol molecules relative to the clay surfaces. Since paracetamol exists in its neutral form at pH 5 (Fig. 5), cation exchange is precluded, as the molecule carries no net charge. The fact that paracetamol binds effectively even when interlamellar sites are blocked confirms that sequestration is restricted to the external clay edges.

The independent t-Student test ($t = 2.446 < t_{\text{crit}} = 2.776$, $\text{df} = 4$) confirmed that there is no statistically significant difference between the two clay systems. This statistical evidence strongly supports the hypothesis of edge-site binding, likely through specific interactions with protonated aluminol groups at the mineral edges. The lack of statistical difference confirms that the expansion of the interlayer spacing from $11.95\text{ \AA}$ to $17.53\text{ \AA}$ (Fig. 2) does not provide additional accessible sites, reinforcing the edge-site binding model.

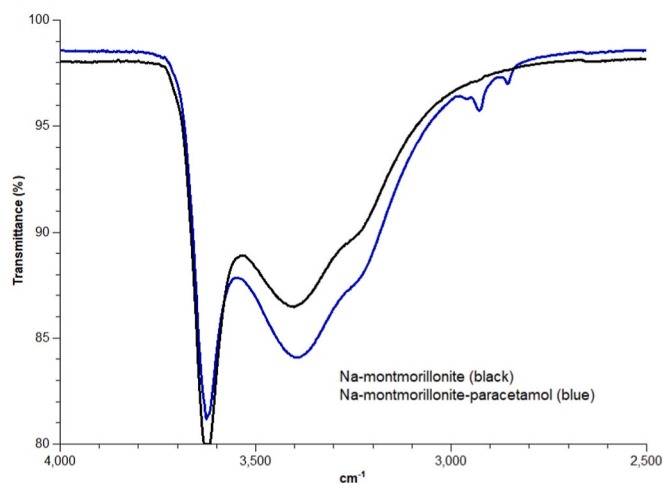


Fig. 7. Comparative ATR-FTIR spectra of natural Na-montmorillonite (black) and the paracetamol-montmorillonite complex (blue) obtained at pH 5. The immobilization of paracetamol on the mineral surface is evidenced by the appearance of distinctive methyl group vibrations at 2926 and 2854 cm^{-1} , which are absent in the natural clay. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1

ATR-FTIR vibrational assignments for clay-adsorbate complexes.

Functional group	Vibration	Frequency (cm^{-1})	Observations after adsorption
Structural OH (Silicate layers)	Stretching (Si-OH)	3640	Characteristic band of the clay matrix
Hydration (Si-OH)	Bending/Hydration	1633	Related to water adsorbed on silicate sites
-CH ₃ (HDTMA-Clay)	Bending/Stretching	2925, 1469, 1380	Evidence of successful organic modification
-CH ₃ (paracetamol)	Symmetric/Asymmetric Stretching	2926, 2854	Indicates successful surface interaction

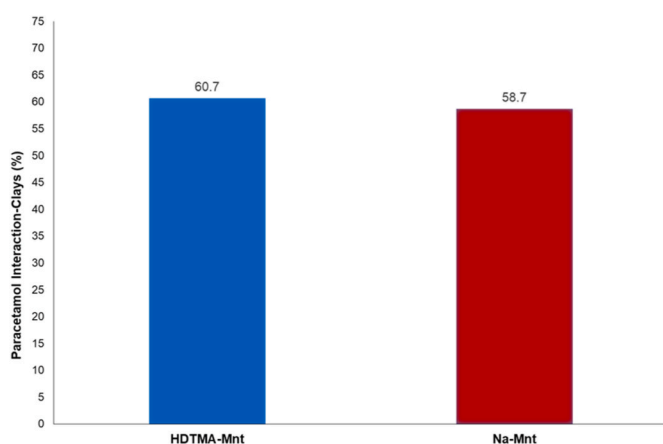


Fig. 8. Comparative adsorption efficiency of paracetamol on Na-montmorillonite (red) and channel-blocked HDTMA-montmorillonite (blue) at pH 5. The statistically equivalent uptake confirms that sequestration is restricted to the reactive mineral edges, as the bulky surfactant cations effectively inhibited access to the interlayer channels. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.4. Alkaline stability tests

Following the pH 5 and 9 experiments, the dried pre-adsorbed paracetamol-clays were treated in an alkaline aqueous medium (pH 12). The processes involved shaking the paracetamol-clay with KOH for different time intervals (15, 30, and 60 min). UV-Vis spectrophotometry and ESI-HPLC-MS confirmed the chemical identity of paracetamol after desorption. It was difficult to desorb the paracetamol from the clay, and the alkaline medium did not favor the desorption process.

3.5. Ionizing irradiation experiments

The radiolytic behavior of paracetamol was evaluated across the different experimental environments to quantify the influence of the mineral matrix on its degradation kinetics and stability. Fig. 9 illustrates the comparative removal efficiencies as a function of the absorbed dose for the aqueous, suspension, and solid-state systems.

3.5.1. Aqueous paracetamol as a homogeneous control

To establish a baseline for degradation efficiency, air-saturated aqueous solutions ($15 \mu\text{g mL}^{-1}$, initial pH 5) were evaluated. In this homogeneous environment, paracetamol is highly labile, undergoing rapid degradation and reaching near-complete removal at low doses ($\leq 1.5 \text{ kGy}$), as illustrated in Fig. 9. According to Yang et al. (2008), this high efficiency is primarily driven by the attack of hydroxyl radicals ($\bullet\text{OH}$) generated through water radiolysis, where the primary yield is $G(\bullet\text{OH}) = 0.28 \mu\text{mol J}^{-1}$. Under the air-saturated conditions of this study, oxidative species such as $\bullet\text{OH}$ and $\bullet\text{HO}_2$ are the principal agents responsible for the breakdown of the aromatic structure.

The radiolytic transformation products were experimentally verified via ESI-HPLC-MS analysis of aqueous phase after exposure 1 kGy (Fig. 10). The identification of key transformation products, specifically hydroquinone ($m/z = 109.04/108.08 [\text{M}-\text{H}]^-$), acetamide ($m/z = 58.97$), and malonic acid ($m/z = 105.96$) provides direct evidence that confirms the degradation products previously reported by Yang et al. (2008); Szabó et al. (2012) (illustrated in Fig. 11).

According to them, hydroquinone formation proceeds through an unstable para-adduct followed by deamination and the simultaneous release of acetamide. While these intermediate by products remain

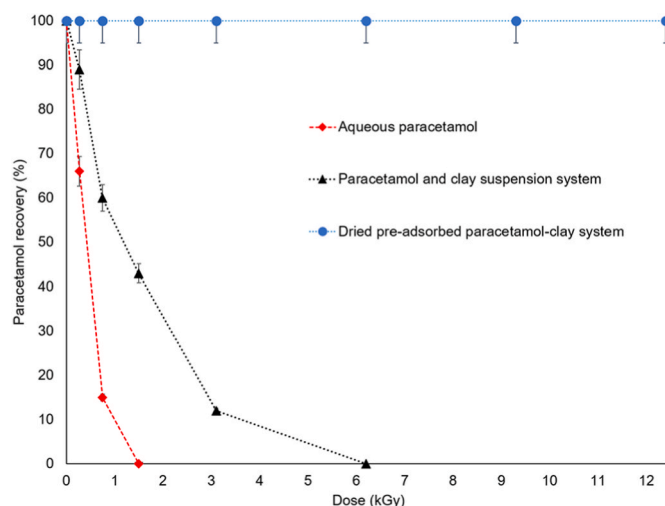


Fig. 9. Removal efficiency of paracetamol as a function of the absorbed gamma dose in three configurations: aqueous phase (red and $\diamond$), clay-water suspension (black and $\blacktriangle$), and dried pre-adsorbed complex (blue and $\bullet$). The data highlight the contrast between rapid mineralization in water and the pronounced radiation resistance induced by the mineral framework in the solid state. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

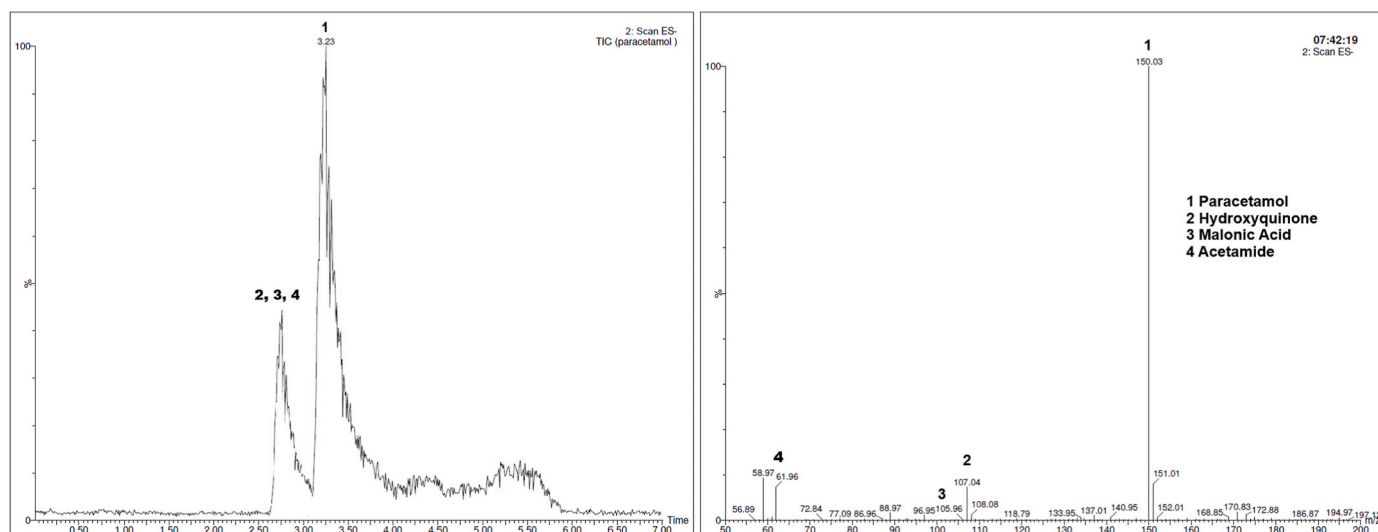


Fig. 10. ESI-HPLC-MS analysis of paracetamol in aqueous solution after exposure to an absorbed dose of 1 kGy. Left: Chromatogram showing the resolution of paracetamol (1) and its radiolytic by-products (2, 3, 4). Right: Negative-ion ESI mass spectrum highlighting the molecular identification of: (1) paracetamol (m/z 150.03), (2) hydroxyquinone (m/z 109.04), (3) malonic acid (m/z 105.96), and (4) acetamide (m/z 58.97). Formic acid, although proposed in the mechanism, could not be identified due to analytical limitations (volatility and mass range).

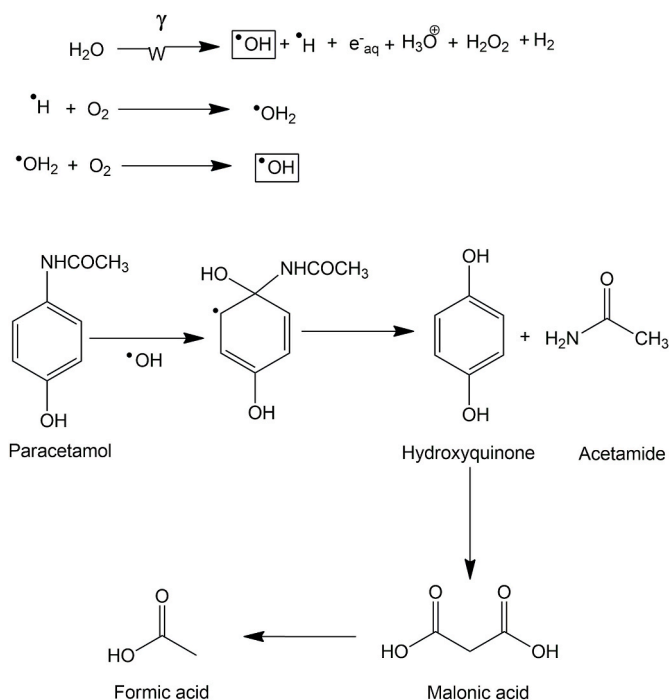


Fig. 11. Schematic representation of the radiolytic degradation of paracetamol in air-saturated aqueous solutions. The process is primarily driven by the attack of hydroxyl radicals ($\bullet\text{OH}$) generated through water radiolysis, which leads to aromatic ring opening. Adapted from Yang et al. (2008).

susceptible to continued radical attack, they eventually lead to aromatic ring opening and the formation of low-molecular-weight carboxylic acids, such as malonic acid, as the system approaches mineralization. The generation of these acidic species is consistent with the observed decrease in the pH of the samples, which dropped from 5 to 4.5 following irradiation. It is important to note that while formic acid is proposed as a terminal intermediate in the schematic mechanism, it could not be identified under the current analytical conditions due to its low molecular weight and high volatility.

3.5.2. Paracetamol and clay suspension systems

In paracetamol-clay suspensions, the degradation follows heterogeneous radiolytic pathways primarily dictated by the specific interaction at the mineral-water interface. At pH 5, paracetamol is selectively immobilized at the clay edges through selective interaction with reactive edge sites, a state that significantly reduces the accessibility of water-derived radicals (e.g., $\bullet\text{OH}$) to the paracetamol. Despite this hindered accessibility, experimental evidence obtained via ESI-HPLC-MS indicates that the degradation intermediates identified in the presence of the mineral (e.g., hydroxyquinone, acetamide, and malonic acid) are identical to those found in the bulk aqueous phase. The detection of these identical products confirms that the fundamental degradation mechanism remains unchanged in the heterogeneous system.

This indicates that while the aluminosilicate matrix acts as a protective barrier that reduces the overall degradation efficiency, it does not alter the underlying radiolytic pathways. Rather than directly participating in the chemical transformation of the pharmaceutical, the mineral matrix functions as a modulator of the reaction microenvironment, providing effective matrix-induced protection. Consequently, while these macroscopic experiments do not provide direct mechanistic proof, the significant decrease in radiolytic efficiency is proposed to be governed by two main phenomena. First, the clay particles may act as competitive scavengers, driving competition kinetics for hydroxyl radicals $\bullet\text{OH}$ between the adsorbed paracetamol molecules and the reactive edge sites. Second, the aluminosilicate lattice likely absorbs and preferentially dissipates the incident ionizing energy, thereby reducing the effective radiation dose available to destroy the pharmaceutical. Ultimately, in agreement with Ramos-Bernal and Negrón-Mendoza (1998), the mineral matrix functions as a profound modulator of the reaction microenvironment.

A robust approach to evaluate the global effect of this complex system is through the initial radiolytic yield $G_0(\text{Para})$, as it captures the net macroscopic degradation efficiency. As shown in Fig. 12, there is a drastic reduction in the effective radiolytic yield, which drops from G_0 of 1.07 (molecules/100 eV) in the bulk water system to 0.37 in the water-clay suspension, and completely to 0 for the dry solid.

The significant decrease of G_0 indicates that the clay matrix exerts a strong protective effect over the pollutant, rendering paracetamol considerably more resilient in complex environmental matrices such as anhydrous soil. This matrix-induced protection suppresses the radiolytic breakdown of the adsorbed species, likely through a mechanism where

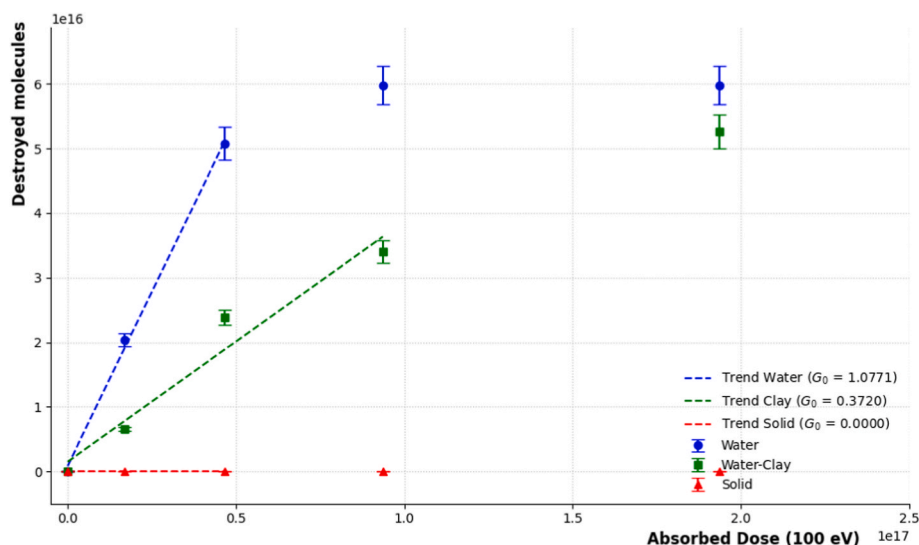


Fig. 12. Number of destroyed paracetamol molecules as a function of the absorbed dose (100 eV) to determine the initial radiolytic yield (G_0) in different microenvironments. The plot compares the degradation efficiency in the bulk aqueous phase (blue and ●), a heterogeneous paracetamol-clay suspension (green and ■), and the dried solid state (red and ▲). The dashed lines represent the linear fits applied to calculate the respective G_0 values. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

ionizing energy is associated with localized structural alterations of the mineral framework rather than the decomposition of the pharmaceutical.

3.5.3. Dried pre-adsorbed paracetamol–clay systems

The high radiation stability of the dried paracetamol–clay system was confirmed through a ATR-FTIR and DSC/TGA techniques and approach following exposure to high-dose gamma radiation.

ATR-FTIR spectra provided primary evidence of the pharmaceutical's structural integrity; even at the maximum dose of 12.4 kGy, characteristic vibrations of the aromatic ring (1510 cm^{-1}) and methyl groups (2926 and 2854 cm^{-1}) persisted without significant shifts. This stability is intrinsically linked to the molecular confinement at the mineral edges, which significantly reduces the accessibility of radiolytic radicals to the pharmaceutical structure. As established by the channel-blocking experiments and the independent t-Student test, paracetamol remains selectively immobilized via inner-sphere complexes with protonated aluminol groups rather than being intercalated. This binding strength is further validated by the complex's resistance to desorption even under extreme alkaline conditions (pH 12), proving the irreversible nature of the edge-site interaction.

The evidence suggests that the mineral framework modulates the reaction microenvironment by undergoing a degree of structural compromise to preserve the chemical integrity of the sequestered pharmaceutical. This protective mechanism, where ionizing energy causes localized framework alterations rather than the breakdown of the adsorbed paracetamol, is characterized by the attenuation of the structural Si-O vibrations ($\sim 1000\text{ cm}^{-1}$) (Fig. 13, black line vs. blue line) in the post-irradiated solid phase (Cruz-Castañeda et al., 2026; Rubino et al., 2024).

ATR-FTIR evidence was further supported by DSC/TGA analysis (Fig. 14), which quantified the organic load retention. The thermograms revealed three distinct weight loss stages: dehydration at 92°C , a critical second peak at 169°C corresponding to the thermal degradation of paracetamol, and lattice dehydroxylation at 700°C . The persistence of the 169°C signature post-irradiation confirms that paracetamol remains a stable organic constituent within the mineral matrix.

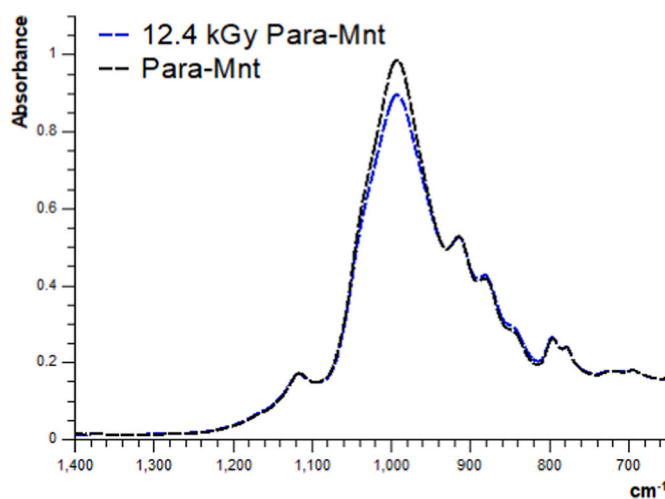


Fig. 13. Comparative solid-state ATR-FTIR spectra of the paracetamol–montmorillonite complex before (0 kGy, black line) and after exposure to a high absorbed dose of 12.4 kGy (blue line). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.6. Radiolytic fate of pharmaceuticals in the soil mineral fraction: the role of the mineral matrix in radiolytic inhibition

The pronounced radiation stability conferred by the montmorillonite matrix to paracetamol highlights a critical environmental paradox: while clays are effective for the sequestration of pollutants, they simultaneously function as protective shields that promote the long-term persistence of emerging contaminants in terrestrial environments. The results demonstrate that molecular confinement at the mineral edges transforms paracetamol from a highly labile molecule in the aqueous phase into a recalcitrant species capable of surviving ionizing radiation doses up to 12.4 kGy, a threshold far exceeding the levels required for its complete mineralization in bulk water. This stability is fundamentally linked to molecular confinement at the mineral interface, which suppresses the degradative capacity of water radiolysis products.

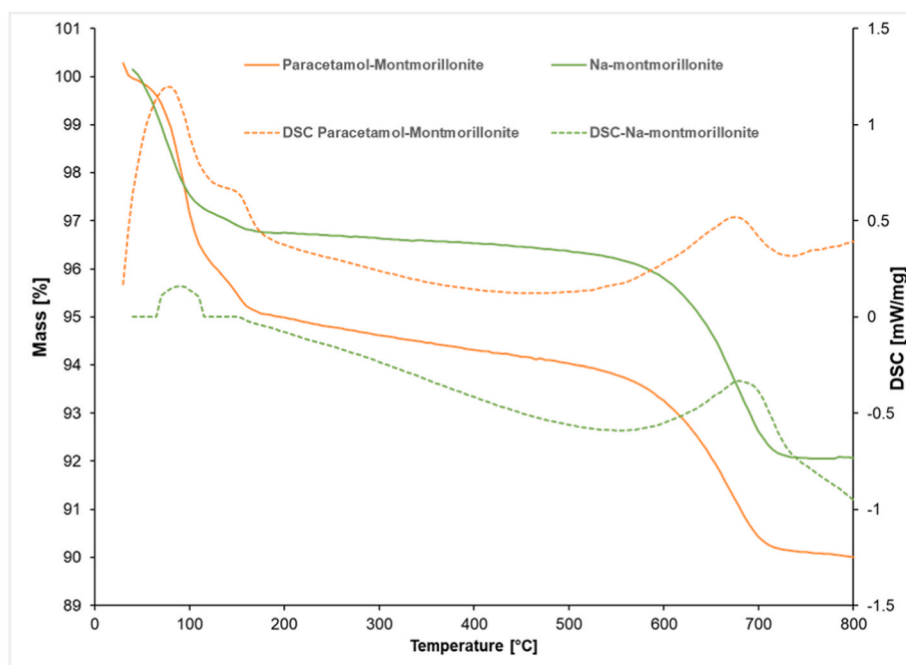


Fig. 14. Simultaneous TGA (solid line) and DSC (dashed line) thermograms for the paracetamol-montmorillonite complex (orange) and natural Na-montmorillonite (green). The thermal signature at 169 °C corresponds to the degradation of the adsorbed pharmaceutical, providing a quantitative basis for the organic load remaining in the system. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

This mineral-induced recalcitrance reveals that clay-rich soil horizons may act as formidable reservoirs where pharmaceuticals remain chemically intact, resisting both natural attenuation and engineered advanced oxidation processes. The drastic reduction of $G_0(\text{Para})$ from 1.07 to 0.37 (molecules/100 eV) underscores that the mineral fraction does not merely serve as a passive support but as a modulator of the radiolytic fate (Table 2). Consequently, these findings pose a significant challenge for environmental management, as the stabilization of toxic residues within the mineral matrix could lead to the accumulation of a latent chemical load in soils, necessitating the design of specialized remediation strategies to overcome the physical protection provided by the mineral framework.

4. Conclusions

This study establishes that the radiolytic stability of paracetamol is fundamentally governed by its interaction with the soil mineral fraction, where the montmorillonite matrix functions as a modulator of the

reaction microenvironment. While paracetamol in the bulk aqueous phase undergoes rapid radiolytic transformation, its association with the aluminosilicate framework confers a pronounced degree of stability. The quantitative impact of this protective effect is demonstrated by the initial radiolytic yield of paracetamol ($G_0(\text{Para})$), which exhibits a 65.5% reduction, decreasing from 1.07 molecules/100 eV in the aqueous system to 0.37 molecules/100 eV in the clay-water suspension. Notably, the pharmaceutical remains chemically intact up to the maximum evaluated dose of 12.4 kGy in the solid-state dried complex, as corroborated by the persistence of its characteristic thermal signature at 169 °C.

A fundamental finding of this work is that the mineral matrix does not alter the radiolytic transformation pathway but significantly suppresses its efficiency. Experimental evidence from ESI-HPLC-MS analysis identified identical intermediates in both aqueous and clay-water systems, confirming that the degradation mechanism remains consistent with the established literature. This stabilization is attributed to molecular confinement at reactive edge sites, statistically validated through interlamellar channel-blocking experiments, where the clay framework potentially restricts the accessibility of water radiolysis products through likely competition kinetics for $\bullet\text{OH}$ radicals.

Furthermore, spectroscopic evidence suggests that this matrix-induced protection is linked to the ability of the aluminosilicate lattice to absorb and dissipate ionizing impact through its own structure, as evidenced by the attenuation of structural Si-O vibrations ($\sim 1000\text{ cm}^{-1}$). Consequently, clay-rich soil horizons could promote the long-term persistence of emerging contaminants by acting as protective environmental reservoirs. Although formal field studies are still required to definitively prove this long-term phenomenon, these results highlight a significant challenge for environmental remediation strategies in terrestrial environments.

CRedit authorship contribution statement

A.S. Loza-Vega: Writing – review & editing, Methodology, Formal analysis. **J.A. Cruz-Castañeda:** Writing – review & editing, Investigation, Formal analysis, Data curation. **A. Negrón-Mendoza:** Writing –

Table 2

Comparison of radiolytic yields $G_0(\text{Para})$ and degradation efficiency of paracetamol in aqueous and clay systems.

System environment	Radiolytic yield $G_0(\text{Para})$ (molecules/100 eV)	Max. Dose evaluated (kGy)	Observed paracetamol radiolytic behavior
Bulk Aqueous Phase (Air-saturated)	1.07	≤ 1.5	Rapid degradation mediated by unrestricted radical diffusion
Aqueous Suspension (Paracetamol-Montmorillonite)	0.37	≥ 6.5	Reduced radical accessibility due to molecular confinement at edge sites of clay
Dried Pre-adsorbed Complex	≈ 0	> 12.4	Marked stability; clay acts as a modulator of the radiolytic fate

review & editing, Investigation, Funding acquisition. **A.L. Meléndez-López:** Writing – original draft, Supervision, Resources, Methodology, Data curation, Conceptualization.

Declaration of AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Google Gemini for English language editing and stylistic refinement. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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